

Kerma - New Task 1

1 Instructions

Each team should submit a report that explains *how your node handles the scenarios listed below*. For each scenario, briefly describe the rationale behind your solution and include a small code snippet that shows the relevant part of your (or the skeleton's) implementation. You don't have to prove anything formally, only identify which parts of the code capture the behaviour described by the following scenarios.

We **highly encourage** you to implement and test your own code, as it will help with future tasks. However, code submissions will **not** be counting towards your grade for Task 1. Nevertheless, If you choose to submit your own implementation, we will provide feedback (including the grader results). We are always available to help out and answer any question.

In the following assume that Grader is a node interacting with your node.

- Grader should receive a valid hello message on connecting.
- The hello message should be followed by a getpeers message.
- Grader should be able to disconnect, then connect to your node again.
- If Grader sends a getpeers message, it must receive a valid peers message.
- If Grader sends {"type":ge, waits for 0.1 second, then sends tpeers"}, your node should reply with a valid peers message.
- If Grader sends any message before sending hello, your node should send an error message with name set to INVALID_HANDSHAKE and then disconnect.
- If Grader sends an invalid message, your node should send an error message with the correct name. Some examples of invalid messages are:
 1. Wbgygvf7rgtyv7tfbgy{{{
 2. "type":"diufygeuybhv"
 3. "type":"hello"
 4. "type":"hello","version":"jd3.x"
 5. "type":"hello","version":"5.8.2"
- If Grader sends a set of peers in a valid peers message, disconnects, reconnects and sends a getpeers message, it must receive a peers message containing (at least) the peers sent in the first message.

- Grader should be able to create two connections to your node simultaneously.

How to Submit Your Report

Please upload a single PDF per group to the corresponding TUWEL submission. Name your file exactly as follows:

report-<group_number>.pdf

How to Submit your Solutions

If you choose to submit your own solution, we will evaluate your solution locally and give you feedback. Please upload your submission as a tar archive. When grading your solution, we will perform the following steps:

```
tar -xf <your submission file> -C <grading directory>
cd <grading directory>
docker-compose build
docker-compose up -d
# grader connects to localhost port 18018 and runs test cases
docker-compose down -v
```

When started in this way, there should be neither blocks (except the genesis block) nor transactions in the node's storage.

If you use the skeleton templates, you can use the provided Makefile targets to build and check your solution. Note however, that this will only check if the container can be built, started and a connection can be established to localhost:18018. We highly recommend that you write your own test cases.

The deadline for this task is 31st October, 11.59pm. We will not accept any submissions after that.